

Git Workflow Guide for Interns

This document explains how to properly create feature branches, update them with the latest code, resolve conflicts, and raise Pull Requests (PRs) to the correct branch. It also covers rebasing as an alternative to merging and provides a recommended .gitignore for MEAN stack applications.

1. Branching Workflow

- Always start from the correct integration branch: main-group-a, main-group-b, main-group-c, or main-group-d.
- Do NOT branch from 'main'.
- Create your feature branch using the naming convention: Group-*-feature/feature-name-intern-name.
- Example: Group-A-feature/login-page-john.
- Command: `git checkout main-group-a && git pull origin main-group-a && git checkout -b Group-A-feature/login-page-john`

2. Keeping Your Branch Updated

- Before raising a PR, always make sure your branch has the latest code from the corresponding main-group-* branch.
- Option 1 (Merge):
 - `git checkout Group-A-feature/login-page-john`
 - `git fetch origin`
 - `git merge origin/main-group-a`
 - Resolve any merge conflicts if they occur, then commit.
- Option 2 (Rebase - cleaner history):
 - `git checkout Group-A-feature/login-page-john`
 - `git fetch origin`
 - `git rebase origin/main-group-a`
 - If conflicts occur, solve them, then run: `git rebase --continue`
 - If you want to abort rebase: `git rebase --abort`

3. Raising Pull Requests

- Always raise PRs to the correct main-group-* branch, NOT to main.
- Title your PR with the feature name and your name.
- Make sure your branch is up to date and conflicts are resolved before raising the PR.

4. Handling Merge Conflicts

- Conflicts occur when two people change the same code lines.
- Git will mark conflicts in files with >>>>>> and <<<<<< markers.
- You must edit the file to keep the correct code.
- After resolving conflicts:
 - `git add`
 - `git commit` (if merging)
 - `git rebase --continue` (if rebasing)

5. .gitignore for MEAN Stack

- node_modules/
- dist/
- build/
- coverage/
- package-lock.json (optional, if lock file not needed)
- environment.ts and environment.prod.ts (if they contain secrets)
- .env
- logs/
- *.log
- .DS_Store
- Thumbs.db

Following these practices will ensure smooth collaboration, clean Git history, and fewer merge conflicts.